



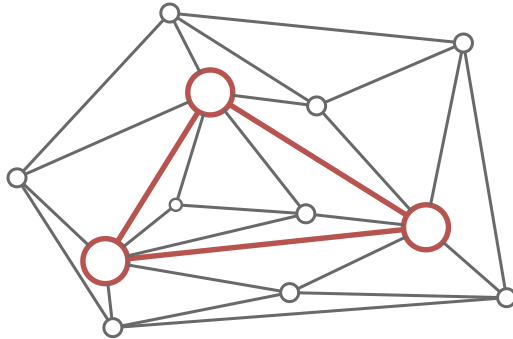
Technische
Universität
Braunschweig



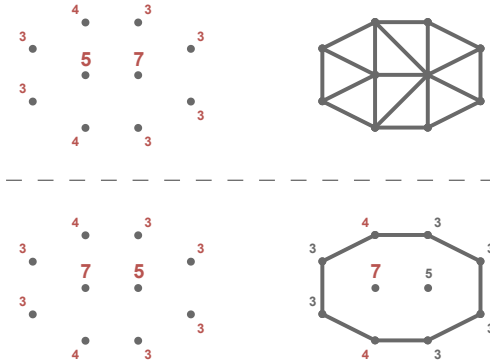
Algorithmische Methoden zu gradbeschränkten Triangulierungen

Fabian Alich, 2. September 2025

Praktische Relevanz



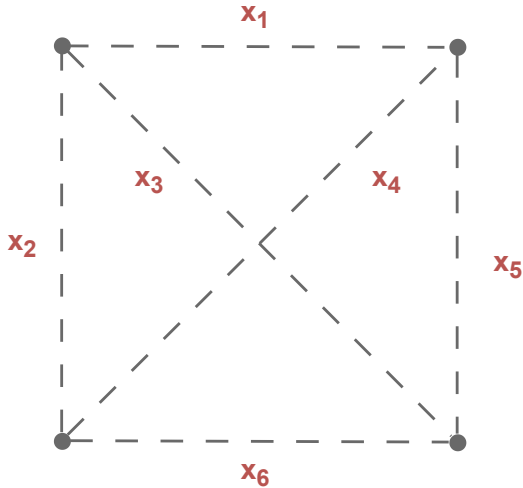
Das Problem



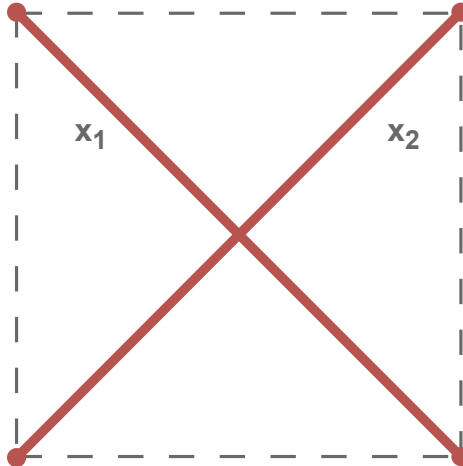
Teil I

Lösungsansätze

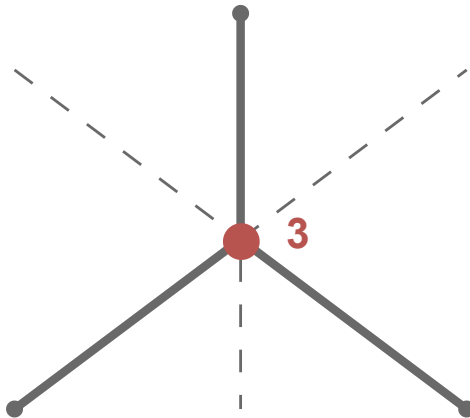
Kantenansatz



Überschneidungsbedingung



Gradbeschränkung



Gradbeschränkung

Durchschnitt

Gradbeschränkung

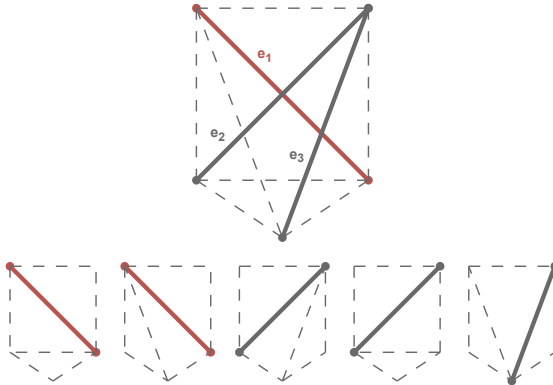
Max min

Maximale Kanten

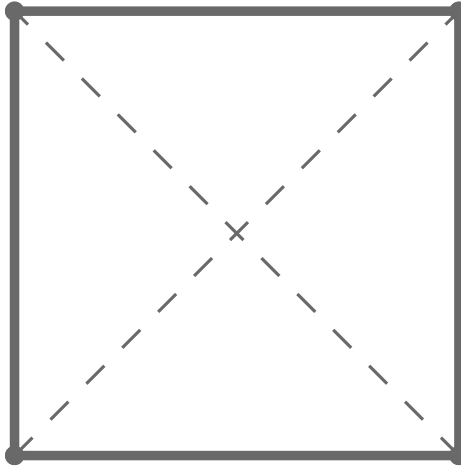
FA: Formal raus?

$$\text{Anzahl Kanten} \cdot 2 \stackrel{!}{=} \sum \text{Sollgrad}$$

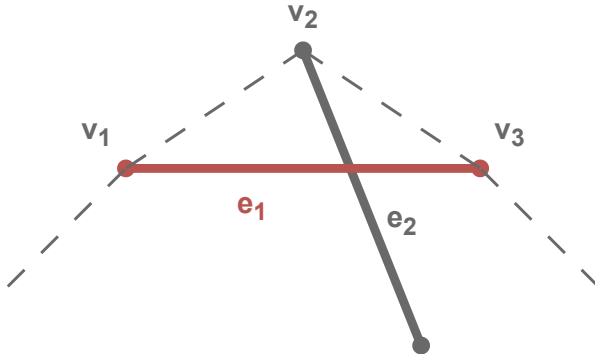
alternative Kantenbedingung



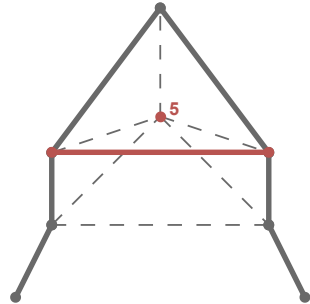
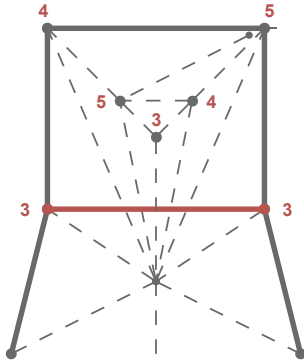
Hülle Fixieren



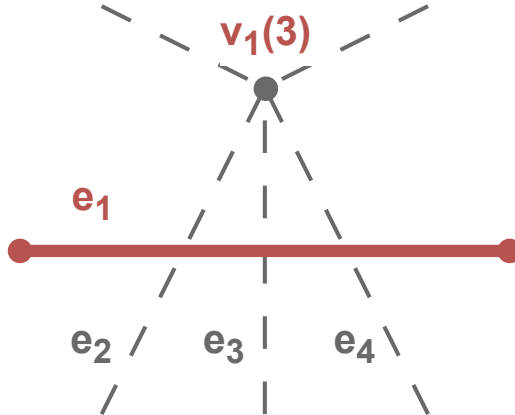
Kanten Fixieren / Ausschließen



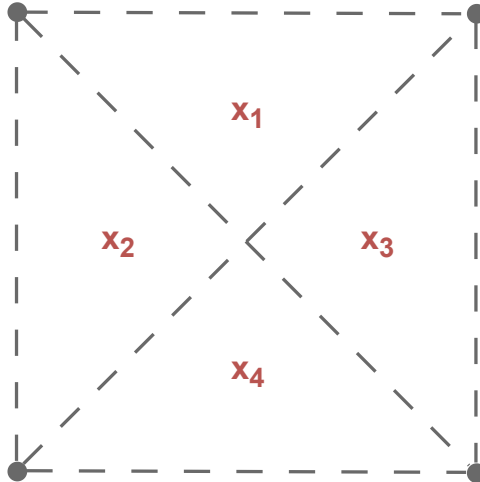
Kanten Ausschließen



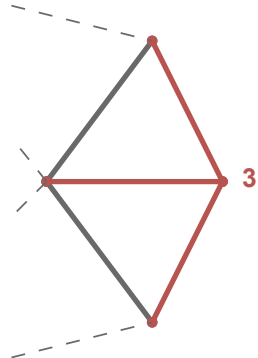
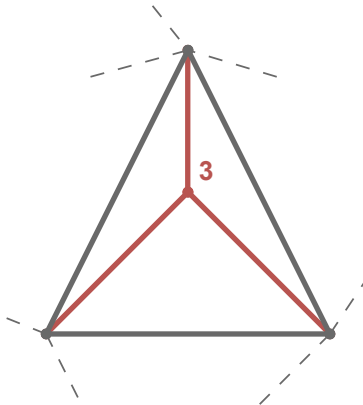
Kanten Ausschließen



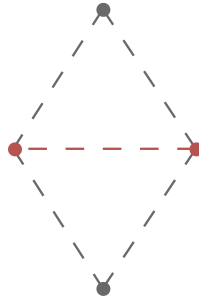
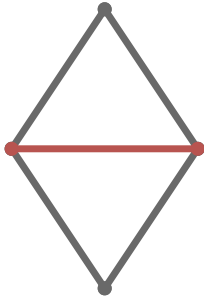
Dreiecksansatz



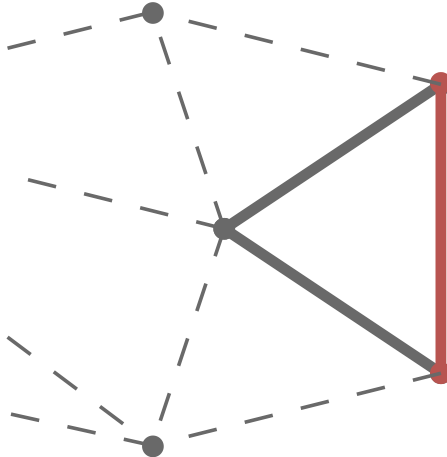
Gradbeschränkung



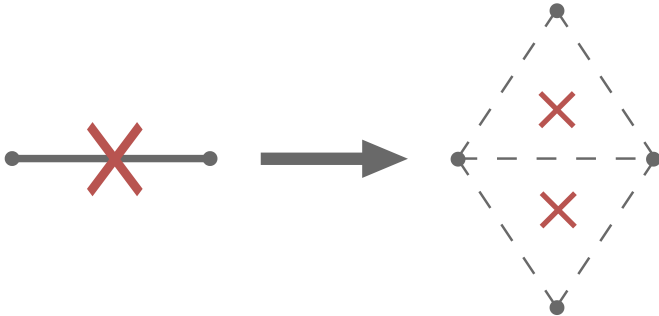
Überschneidungsbedingung



Überschneidungsbedingung



Dreiecks Ausschließen



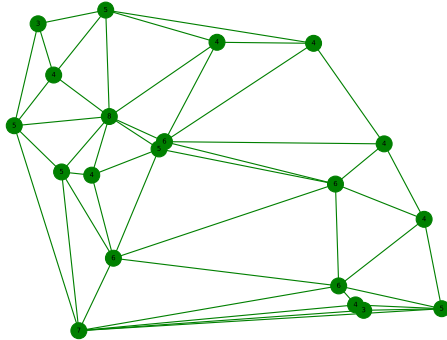
Subset

Soll ich subset erklären oder weglassen?

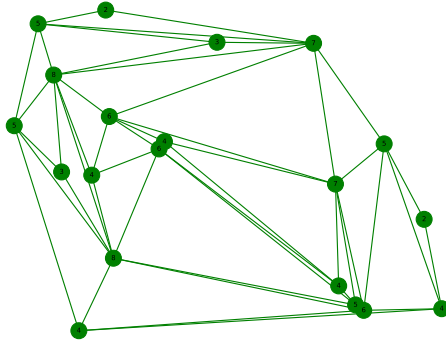
Teil II

Instanzgenerierung

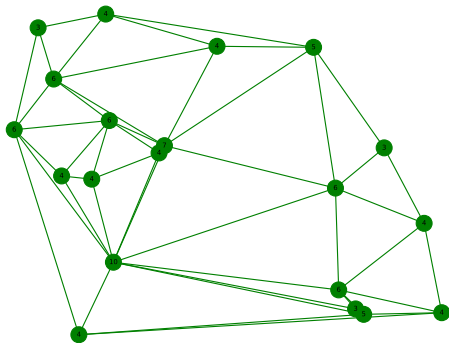
Delaunay



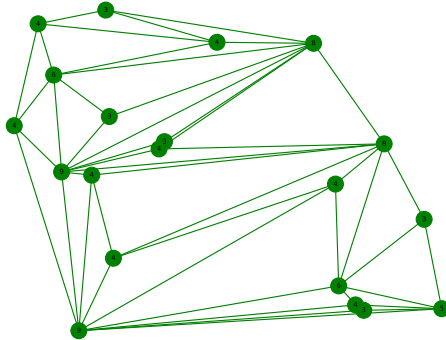
Delaunay-Flips



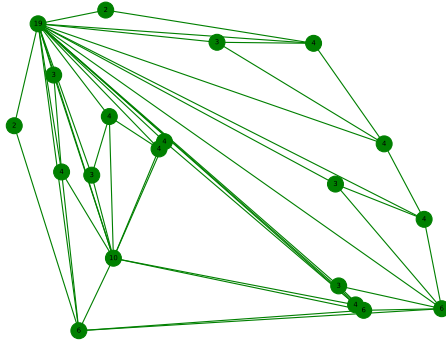
Greedy



Random



Iterativ



Teil III

Auswertung

Solver



Gurobi



OR-Tools



PySAT

Permutation